

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks Available |     |     |       |       |      |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6.       | (a) | (i)  |  | Reduction in units = $15\,000 - 3\,000 = 12\,000$ [kWh] (1)<br>Reduction in cost = $12\,000 \text{ ecf} \times 20 = 240\,000$ [p] (1)<br>Conversion to £ = [£]2 400 (1)<br><br><b>Alternative:</b><br>$15\,000 \times 20 = 300\,000$ [p] (1)<br>$3\,000 \times 20 = 60\,000$ [p] (1)<br>Saving = $3\,000 \text{ ecf} - 600 \text{ ecf} = [£]2\,400$ (1)<br><br>Award 2 marks for answers of $2.4 \times 10^n$ where $n$ is not 3 |                 | 3   |     | 3     | 3     |      |
|          |     | (ii) |  | Payback time = $\frac{152\,400}{2\,400 \text{ ecf}} = 63.5$ [months] accept 64 [months]                                                                                                                                                                                                                                                                                                                                          |                 | 1   |     | 1     | 1     |      |
|          | (b) | (i)  |  | Selection and substitution into efficiency equation:<br>$\frac{1.9}{5.3}(1) [\times 100]$ (1)<br>% efficiency = 35.849 <b>or</b> 35.8 <b>or</b> 35.85 <b>or</b> 36 <b>or</b> 40 (1)<br>Award 1 mark for 0.36 or 0.4                                                                                                                                                                                                              |                 | 2   |     | 2     | 2     |      |
|          |     | (ii) |  | Weekly mass of dung = $150 \times 60 = 9\,000$ [kg/week] (1)<br>Weekly volume of gas = $9\,000 \text{ (ecf)} \times 0.05 = 450$ [m <sup>3</sup> ] (1)<br>Weekly number of kWh = $450 \text{ (ecf)} \times 1.9 = 855$ [kWh] (1)<br>Annual number of kWh = $855 \text{ (ecf)} \times 52$ (1)<br>= 44 460 <b>(ecf)</b> [kWh] [so farmer is incorrect] (1)                                                                           |                 |     | 5   | 5     | 5     |      |